

15BMO

SomeonecoolLovesMaths

June 9, 2026

Problem

Let $ABCD$ be a cyclic quadrilateral. Let F be the midpoint of the arc AB of its circumcircle which does not contain C or D . Let the lines DF and AC meet at P and the lines CF and BD meet at Q . Prove that the lines PQ and AB are parallel.

Solution. Since F is the midpoint of arc AB , we have $\angle BCF = \angle FCA = \angle FDA = \angle FDB$. Since $B - Q - D$ and $A - P - C$, we know that $\angle PDQ = \angle PCQ$ which means that $PQCD$ is cyclic. So $PQ \parallel AB$ due to Reim's Theorem.